The Hall Effect in a Two-Dimensional Electron Gas

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Abstract

This report uses representative data, generated to demonstrate the analysis rather than measured by the author; the note below sets out what that means. On that basis, the Hall coefficient and the resistance per square of a GaAs/AlGaAs modulation-doped heterostructure containing a two-dimensional electron gas were determined on a six-contact Hall bar of three squares, at room temperature and under liquid nitrogen cooling. An iron-cored electromagnet was calibrated against a Hall probe magnetometer with a cubic polynomial, a straight line overreading the adopted cubic at the top of the sweep by 14 mT, 4 per cent of the calibrated field there. The Hall voltage was shown to be a single linear function of the product of sample current and normal flux density, and its variation with the angle between the bar and the field followed $V_{H0}\cos(\theta-\theta_0)+C$ with $\theta_0 = (6.07 \pm 0.08)^\circ$. A three-parameter global weighted fit separating the Hall term from a field-independent contact-misalignment term gave $R_{H,2\mathrm{D}} = (1235 \pm 48)\,\Omega\,\mathrm{T}^{-1}$ at 300 K and $(1406 \pm 54)\,\Omega\,\mathrm{T}^{-1}$ at (79 ± 3) K, while zero-field longitudinal sweeps gave $R_{\mathrm{sq}} = (1561 \pm 16)\,\Omega/\square$ and $(117.9 \pm 1.2)\,\Omega/\square$. The sheet carrier density fell from $(5.05 \pm 0.20) \times 10^{15}\,\mathrm{m}^{-2}$ to $(4.44 \pm 0.17) \times 10^{15}\,\mathrm{m}^{-2}$, a change of -12.2 per cent, while the mobility rose from $(0.791 \pm 0.032)\,\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ to $(11.93 \pm 0.48)\,\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, a factor of 15.08 ± 0.04 in which the magnet and geometry systematics cancel, so that uncertainty is statistical only. Both mobilities agree with published values for comparable structures to within 3 per cent. The contrast between an almost constant carrier density and a mobility that rises by more than an order of magnitude is the expected fingerprint of modulation doping.

Note on the data in this report. The numbers, tables and figures in this report are *representative data*. They were generated from physically accepted parameters with realistic instrument scatter and realistic systematic offsets added, and then analysed exactly as laboratory readings would have been. They are *not* a record of measurements taken by the author and must not be read as one. The original laboratory notebook is no longer available, so the report has been written up from the departmental experiment information script in order to demonstrate the method, the analysis and the treatment of uncertainty. Every results table and every data figure says "representative data" in its caption, and every page repeats the statement in its footer.

1 Introduction

The Hall effect has been the standard way of counting charge carriers in a conductor since Hall's experiment of 1879 [2]. A current is passed along a conductor and a magnetic flux density is applied normal to it. The Lorentz force pushes carriers to one side, charge accumulates until the transverse electric field it creates cancels that force, and the resulting transverse voltage gives the carrier density directly, with its sign giving the sign of the carrier charge. Combined with a measurement of the resistivity it also gives the carrier mobility, which is the most useful single figure of merit for a semiconductor material.

The sample here is not a bulk conductor. It is a GaAs/AlGaAs heterostructure in which electrons are trapped in a narrow potential well at the interface between two semiconductors of different band gap. They move freely in the two dimensions parallel to the interface but are confined normal to it, forming a two-dimensional electron gas (2DEG). Because the donors that supply the electrons sit in a different layer from the electrons themselves, a scheme called modulation doping [5], the

carriers are separated from the ionised impurities they came from and the mobility is far higher than any uniformly doped material can reach [6]. That is the property exploited in the high electron mobility transistor, which uses this layer structure as its channel and is the standard low-noise front-end device at microwave and millimetre-wave frequencies [3]. Mobility and sheet density at two temperatures, the measurement made here, are the two numbers that characterise such a layer structure.

Following the objectives of the departmental script [1], the experiment measures the resistance per square of the 2DEG at room temperature; verifies that the Hall voltage varies with flux density and with sample current in the way the theory predicts, and obtains the two-dimensional Hall coefficient; finds the sheet carrier density and the mobility; investigates how the Hall voltage varies with the angle between the bar and the field; repeats the resistance per square and the Hall coefficient with liquid nitrogen cooling; and asks how best to reduce a whole array of readings taken over a range of fields and currents into one averaged result with a sensible error estimate. The last of these is treated as a question in its own right in Section 4.5, because the three reductions compared there do not all give the same answer.

2 Theory

2.1 The Hall effect in three dimensions

Consider a bar carrying a current I along $\hat{x}$, of width w along $\hat{y}$ and thickness t along $\hat{z}$, with flux density $\mathbf{B} = B\hat{z}$ normal to the wide face. The carriers are electrons of charge $q = -e$ and volume density n. A carrier of drift velocity $\mathbf{v}$ feels the Lorentz force

$$\mathbf{F} = q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B})\,. \tag{1}$$

With $\mathbf{v} = v_x\hat{x}$ and $\mathbf{B} = B\hat{z}$ the vector product is $\mathbf{v} \times \mathbf{B} = -v_x B\,\hat{y}$, so the magnetic force acts across the bar. The carriers cannot leave the sides, so charge builds up on one edge and is depleted from the other, and the transverse field this creates grows until the net transverse force vanishes:

$$qE_y - qv_xB = 0 \qquad \Longrightarrow \qquad E_y = v_xB. \tag{2}$$

This steady state is reached in a time of order the dielectric relaxation time, far shorter than any measurement here [4]. Writing the current density as $J_x = nqv_x = -nev_x$ gives $E_y = -J_xB/(ne)$. The transverse voltage across the width is the Hall voltage $V_H = E_y w$, and with $J_x = I/(wt)$,

$$R_H \equiv \frac{E_y}{J_xB} = \frac{V_H t}{IB} = -\frac{1}{ne}, \qquad |R_H| = \frac{1}{ne}, \qquad |V_H| = \frac{IB}{net}, \tag{3}$$

where R_H is the Hall coefficient, of units $\mathrm{m^3\,C^{-1}}$. The minus sign is the signature of negative carriers: for holes, $q = +e$, the same argument gives $R_H = +1/(pe)$. Magnitudes are quoted throughout this report, so every Hall coefficient below is listed positive, and the polarity actually observed is reported in Section 4.4. A full Boltzmann treatment replaces the numerator of R_H by a Hall scattering factor r_H, which collapses to unity for a degenerate gas because all the transport then happens at the Fermi energy and the energy averaging that produces r_H disappears [4]. Section 5.5 states how good that assumption is here.

2.2 Reduction to two dimensions

In a 2DEG the electrons occupy a sheet whose thickness is set by the extent of the confined wavefunction and is not a well-defined experimental quantity. The meaningful density is therefore the sheet carrier density $n_{\mathrm{sq}} = \int n(z)\,\mathrm{d}z$, in $\mathrm{m^{-2}}$, which for a uniform slab would be nt. The script states the matching rule for the current: the current per unit area is replaced by the current per unit width [1]. Defining the sheet current density $K = I/w$ in $\mathrm{A\,m^{-1}}$ and repeating the argument above, equation (2) is unchanged and

$$R_{H,\mathrm{2D}} \equiv \frac{E_y}{KB} = \frac{V_H/w}{(I/w)B} = \frac{V_H}{IB} = \frac{1}{n_{\mathrm{sq}}e}, \qquad \text{so} \qquad \boxed{V_H = \frac{IB}{n_{\mathrm{sq}}e} = R_{H,\mathrm{2D}}\,IB} \tag{4}$$

The units of $R_{H,2\mathrm{D}}$ are $\mathrm{m^2\,C^{-1}}$, which is the same as $\Omega\,\mathrm{T^{-1}}$: since $\mathrm{T} = \mathrm{V\,s\,m^{-2}}$, $\Omega\,\mathrm{T^{-1}} = \mathrm{V\,m^2\,A^{-1}\,V^{-1}\,s^{-1}} = \mathrm{m^2\,C^{-1}}$.

Equation (4) is the central result. The Hall voltage of a two-dimensional system depends only on the measured current, the component of flux density normal to the sheet, and the sheet density. It contains no sample dimension at all: not the width, not the length and not any thickness. The width cancels because widening the bar reduces the sheet current density and lengthens the path over which the transverse field is integrated by exactly the same factor. The consequence is that n_{sq} obtained from a Hall measurement carries no systematic error from l, from w or from any assumed thickness. The one geometrical imperfection that does reach it is a displacement of the two Hall arms along the bar, and Section 4.5 shows that leaving it unmodelled biases the Hall coefficient by 3.15 per cent at room temperature.

2.3 Resistance per square, and extracting the mobility

In three dimensions the Drude conductivity is $\sigma = ne\mu$, where μ is the mobility, the mean drift velocity per unit applied electric field. Integrating through the sheet gives the sheet conductance $\sigma_{\mathrm{sq}} = n_{\mathrm{sq}} e \mu$, whose reciprocal is the resistance per square,

$$R_{\mathrm{sq}} = \frac{1}{n_{\mathrm{sq}} e \mu}, \tag{5}$$

the two-dimensional analogue of resistivity and the script's first equation [1]. Ohm's law in two dimensions reads $\mathbf{E} = R_{\mathrm{sq}}\mathbf{K}$, so for a region of length l along the current and width w across it,

$$R_L = \frac{V_L}{I} = \frac{E_x l}{K w} = R_{\mathrm{sq}}\,\frac{l}{w}. \tag{6}$$

The resistance depends on the aspect ratio only, not on the absolute size, so a square of 2DEG of any side length has resistance R_{sq}. The unit is written ohms per square to signal that the quantity is a sheet property. Here the voltage probes are separated by $l = 300\,\mu\mathrm{m}$ on a bar of width $w = 100\,\mu\mathrm{m}$ [1], so $l/w = 3$ exactly and $R_L = 3R_{\mathrm{sq}}$.

Equations (4) and (5) are two equations in the two unknowns. Inverting the first and dividing one by the other,

$$n_{\mathrm{sq}} = \frac{1}{e\,R_{H,2\mathrm{D}}}, \qquad \frac{R_{H,2\mathrm{D}}}{R_{\mathrm{sq}}} = \frac{1/(n_{\mathrm{sq}} e)}{1/(n_{\mathrm{sq}} e \mu)} = \mu. \tag{7}$$

In terms of the two quantities actually fitted, the slope $m_H = \mathrm{d}V_H/\mathrm{d}(IB) = R_{H,2\mathrm{D}}$ of the Hall data and the slope $m_L = \mathrm{d}V_L/\mathrm{d}I = R_L$ of the longitudinal data,

$$n_{\mathrm{sq}} = \frac{1}{e\,m_H}, \qquad \mu = \frac{m_H}{m_L}\cdot\frac{l}{w} = \frac{3\,m_H}{m_L}. \tag{8}$$

Equation (7) also provides a check on units. The units of $R_{H,2\mathrm{D}}/R_{\mathrm{sq}}$ are $(\Omega\,\mathrm{T^{-1}})/\Omega = \mathrm{T^{-1}}$, and since $\mathrm{T} = \mathrm{V\,s\,m^{-2}}$ this is $\mathrm{m^2\,V^{-1}\,s^{-1}}$, exactly the unit of mobility.

2.4 Angular dependence of the Hall voltage

Let the sheet lie in the xy plane with normal $\hat{z}$ and the current along $\hat{x}$. The confinement means the carriers have no velocity component normal to the sheet whatever the field direction, so for a general field $\mathbf{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$,

$$\mathbf{v} \times \mathbf{B} = v_x B_y\,\hat{z} - v_x B_z\,\hat{y}. \tag{9}$$

The three components have different fates. The term in B_x, parallel to the current, vanishes identically. The term in B_y pushes the carriers out of the plane and is opposed by the confining potential of the quantum well, which is steep on the scale of tens of millielectronvolts, so no charge accumulates and no in-plane voltage appears. Only the term in B_z drives charge to the sides of the bar. Only the component of $\mathbf{B}$ normal to the 2DEG therefore produces a Hall voltage. Writing θ for

the angle read from the rotation stage and θ_0 for the reading at which the sample normal is truly parallel to the field, $B_z = B\cos(\theta - \theta_0)$ and

$$V_H(\theta) = V_{H0}\cos(\theta - \theta_0) + C, \qquad V_{H0} = \frac{IB}{n_{\mathrm{sq}} e}. \tag{10}$$

The constant C is not part of the ideal prediction and is added deliberately. It represents a field-independent voltage on the Hall contact pair, whose dominant cause is contact misalignment: if the two Hall arms are displaced by a small distance δ along the current direction rather than lying exactly opposite one another, the pair straddles a short length of the bar and picks up a fraction of the longitudinal voltage,

$$V_{\mathrm{offset}} = I\, R_{\mathrm{sq}}\, \frac{\delta}{w}. \tag{11}$$

This term is proportional to I but independent of B and of angle. Since R_{sq} is much larger at room temperature than under cooling while $R_{H,\mathrm{2D}}$ barely changes, the offset problem should be far worse at room temperature, a prediction tested in Section 6.3.

2.5 Why the electrons are two-dimensional

The sample is a stack grown epitaxially on a thick GaAs substrate: n-AlGaAs doped with donors, an undoped AlGaAs spacer, then intrinsic GaAs [1]. AlGaAs has the wider band gap, so at the interface there is a conduction band offset of roughly 0.22 eV to 0.25 eV for an aluminium fraction near 0.3 [9]. The donors sit at a higher conduction band energy than the GaAs conduction band edge, so electrons lower their energy by transferring across the interface, leaving ionised donors behind. The resulting dipole bends the GaAs conduction band sharply upwards away from the interface, and together with the abrupt barrier on the AlGaAs side this gives an approximately triangular well, roughly 10 nm to 13 nm wide in the GaAs depending on the interface field assumed. Solving the Schrödinger equation for a triangular well of interface field F gives subband energies

$$E_j = \left(\frac{\hbar^2}{2m^*}\right)^{1/3}\left[\frac{3\pi e F}{2}\left(j + \tfrac{3}{4}\right)\right]^{2/3}, \qquad E(j, \mathbf{k}_\parallel) = E_j + \frac{\hbar^2 k_\parallel^2}{2m^*}, \tag{12}$$

so each subband carries a free two-dimensional band of in-plane motion [7]. For the sheet densities found here the estimated spacing between the lowest two subbands is 34 meV to 54 meV, the range spanned by taking the interface field from half the sheet charge or from all of it. At 79 K that spacing is far larger than $k_B T = 6.81$ meV and the second subband is all but empty, holding under 3 per cent of the carriers even at the bottom of the spacing range.

At 300 K the margin is thinner, and the occupation has to be worked out with the right quantity. The Fermi energy of 18.06 meV in Table 8 is a degenerate-gas expression, but at 300 K this gas is not degenerate, $T/T_F = 1.43$, so it is the chemical potential and not E_F that sets the occupation. Filling a subband of constant two-dimensional density of states g_{2D} and edge E_j to a chemical potential μ_c, written with a subscript here so that it is not read as the mobility, gives

$$n_j = g_{\mathrm{2D}}\, k_B T\, \ln\!\left[1 + \exp\!\left(\frac{\mu_c - E_j}{k_B T}\right)\right], \tag{13}$$

which collapses to the degenerate result $n_j = g_{\mathrm{2D}}(\mu_c - E_j)$ only when $\mu_c - E_j \gg k_B T$. Solving equation (13) for the lowest subband alone at the room-temperature sheet density puts μ_c at 0.28 meV above the subband edge, not at 18.06 meV. Solving it for both subbands together, so that $n_0 + n_1 = n_{\mathrm{sq}}$, puts μ_c between -9.4 and -5.0 meV, that is below the lowest subband edge, which is what a non-degenerate gas requires, and leaves 14 to 25 per cent of the carriers in the second subband across the 34 meV to 54 meV spacing range. The single-subband analysis is therefore an approximation at room temperature rather than an exact description, and with up to a quarter of the carriers in the second subband it is not a close one. Any second-subband occupation acts in the same direction as parallel conduction: a second group of carriers of lower mobility in parallel with

the first raises the apparent Hall density above that of the high-mobility channel alone and lowers the apparent mobility.

The undoped spacer displaces the electrons from the donors that supplied them. Scattering from a remote ionised impurity a distance d away is suppressed because the matrix element for scattering through wavevector q carries a factor e^{-qd}, which cuts hardest into the large-angle events that relax momentum most efficiently [8]. Modulation doping therefore breaks the trade-off, normal in a uniformly doped semiconductor, in which raising the carrier density necessarily lowers the mobility. Once phonons are frozen out entirely the mobility becomes a pure measure of wafer purity and can exceed $10^7\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ in modern material [11].

3 Apparatus and method

3.1 Sample, contacts and detection scheme

The Hall bar is defined on the wafer by an etched trench that isolates it electrically, with ohmic contacts at the ends of six arms, and is mounted in a plug-in box carrying a diagram of the bar on its lid. The active region between the voltage probes is $l = 300\,\mu\mathrm{m}$ long and $w = 100\,\mu\mathrm{m}$ wide [1], so it is exactly three squares. Current is driven end to end through a dedicated connector, in series with two $150\,\mathrm{k\Omega}$ protection resistors, and the series current is read on a digital multimeter through a separate pair of ammeter terminals. Contacts 1 to 4 are banana sockets for voltage measurement only and carry no current: the longitudinal voltage V_L is taken between contacts 1 and 4, or 2 and 3, and the Hall voltage V_H between contacts 1 and 2, or 4 and 3.

Both voltage measurements are therefore four-terminal. The current enters and leaves through the end contacts while the voltages are read on separate high-impedance arms that carry no current, so the contact resistances at the current terminals and the protection resistors drop out entirely and the resistance obtained is a property of the $300\,\mu\mathrm{m}$ of 2DEG between the probes and nothing else. The two $150\,\mathrm{k\Omega}$ resistors do not enter the analysis, because the current is measured directly rather than inferred from the supply voltage and the resistor value, so their tolerance contributes nothing to any result. What they do determine is the shape of the experiment: with $300\,\mathrm{k\Omega}$ in series against a sample resistance of a few kilohms the drive behaves as a near-ideal current source, so the current sweep is very linear in supply voltage and the current at a given supply voltage changes very little on cooling.

The longitudinal voltage was measured with the electromagnet switched off. That instruction in the script [1] is not merely a convenience: measuring at zero field removes any residual magnetoresistance, small here because for a single carrier species in a long bar the Drude longitudinal resistance is field-independent and geometric magnetoresistance requires a short, wide sample; it removes any Hall voltage that would leak into the longitudinal pair if contacts 1 and 4 are not exactly in line; and it keeps the magnet calibration out of the error budget for R_{sq} altogether. It matters most at low temperature, where the Hall voltage becomes comparable with, and at the top of the sweep larger than, the longitudinal voltage. Because a field-independent offset of the form of equation (11) is always present on the Hall pair, the analysis fits it explicitly rather than assuming it away.

3.2 Electromagnet and its calibration

The field is produced by an iron-cored electromagnet driven from a laboratory voltage supply, so the only quantity under direct control is the magnet supply voltage. Because the core is ferromagnetic, the relation between coil current and gap field is neither linear nor single valued. The field was therefore measured independently with a calibrated Hall probe magnetometer, with the sample stand removed from the gap, over the full working range and in both directions so that the hysteresis loop could be seen.

Applying Ampère's law around the magnetic circuit gives $N I_m = H_{\mathrm{iron}} l_{\mathrm{iron}} + H_{\mathrm{gap}} l_{\mathrm{gap}}$ with $B = \mu_0 H_{\mathrm{gap}}$. If the relative permeability of the iron were constant, B would be strictly proportional to the coil current I_m. It is not: as the yoke approaches magnetic saturation an increasing share of the magnetomotive force is spent driving flux through the iron rather than across the gap, so the field grows more slowly than linearly at the top of the range and the calibration curve bends over.

The magnet was cycled to maximum current and back before each run and every set point was then approached monotonically from below, using the same sequence of supply voltages in the calibration and in the measurement, so that the field history was at least the same from run to run. It should be said plainly that this does not put calibration and measurement on the same branch. A down-cycle to zero coil current leaves the core at the falling-branch remanence, fitted in Section 4.1 as 8.92 mT, whereas the calibration rising branch starts from 2.56 mT. Ramping up from the remanent state traces a minor loop somewhere between the two branches, and the resulting offset, about 6 mT at low coil current and falling towards zero at the top of the sweep, is carried as a systematic on B in Section 5. Reproducing the rising branch itself would need the core demagnetised before every run, which was not done and is listed among the improvements in Section 6.5.

3.3 The rotation stage and how the angle was set

The plug-in box does not sit loose in the gap. It seats in a holder on a stand that stands between the pole pieces, and the stand turns about a vertical axis, so that the bar rotates in the horizontal plane while the field stays horizontal across the gap. Turning the stand therefore changes the angle between the field and the normal to the 2DEG, which is the quantity equation (10) depends on, and leaves the magnitude of the field alone. The same stand lifts out of the gap for the magnet calibration of Section 4.1, which is why that calibration is taken with the gap empty.

The angle is read against the graduated scale on the stage. It was set and recorded to the nearest degree, which is the resolution at which the scan appears in Appendix F, and what that resolution is worth is worked out in Section 5.2. The zero of the scale was set by eye, by turning the stand until the face of the bar looked square to the pole pieces; no attempt was made to do better than that by eye, because the fit returns the residual misalignment as θ_0 and corrects for it. Before the scan the leads were left slack enough that a full turn would not pull on the plug-in box, since a lead dragging on the box would move the sample as well as turn it.

The scan itself was taken with both supplies held fixed, as the script asks [1]: the magnet supply at 20.5 V and the bar supply at 18.3 V, chosen to give a Hall voltage of a few tens of millivolts, comfortably above the meter scatter and comfortably below the current cap. The stage was then stepped through a full 360° in 15° steps. A full turn was used rather than a half turn for a specific reason: it makes every angle available together with the angle 180° from it, and half the difference of that pair removes the field-independent offset exactly, which is the parity argument used in Section 4.6. The sample current was logged at every angle rather than assumed constant, and it is that log which shows the 0.2 per cent drift discussed there.

3.4 Instrumentation and automated data collection

The bar supply, the magnet supply and the two multimeters are USB instruments and were driven from a Jupyter notebook using the `HallPy_Teach` library [13], which wraps the underlying PyVISA calls. The library is installed into the local Anaconda environment and its setup routine asks the user to confirm the USB serial number of each instrument against the label on its case, so that the notebook knows which meter is reading the Hall voltage, which is reading the longitudinal voltage and which supply is driving the magnet. Getting that assignment wrong is the usual reason for the experiment not to run at all, so it was checked against the wiring before any data were taken. Its `doExperiment` routine takes a list of electromagnet voltages, a tuple giving the lowest and highest bar supply voltage and the number of points per sweep, and returns the sample current with the selected voltage at every point. The library enforces a maximum sample current of 100 µA to protect the bar. For the Hall runs the bar supply was swept from 3.0 V to 27.0 V in 18 steps at each of five magnet settings, giving 90 points per temperature. For the longitudinal runs the magnet voltage list was set to zero and the same bar sweep was used.

The order of work was the order the script sets out [1]. The manual familiarisation readings of Section 3.6 came first, then the magnet calibration with the stand lifted out of the gap, then the room-temperature longitudinal sweep and the room-temperature Hall array, then the angular scan, and only then the cooling and the repeat of the longitudinal sweep and the Hall array. Putting

the calibration before the measurement runs matters, because the field set points used in the measurement have to be the ones the calibration polynomial was fitted through.

3.5 Cryogenic procedure and liquid nitrogen handling

A demonstrator was asked for instructions before this stage was attempted, both for the rotary pump and for the liquid nitrogen itself, which is what the script requires [1]. The sample sits on a cold finger inside a dewar with an evacuated jacket. The jacket was pumped out with the rotary pump before any nitrogen was admitted, so that the cold finger is thermally isolated from the room and the sample space does not frost up from condensed water and air. When the pumping was finished the pump was vented to atmosphere before it was switched off, which is a standing instruction in the script and stops pump oil being drawn back up the line into the dewar.

Liquid nitrogen was then transferred into the dewar. Thermal gloves and goggles were worn for the transfer, and no cooled part of the apparatus, and nothing that had been in contact with the liquid, was touched at any point. Liquid nitrogen boils at 77.4 K at one atmosphere and expands by roughly a factor of 700 when it does, so the transfer was made with the dewar vent clear and in a space with air movement, and the cryogen was never allowed to sit in a closed volume. The sample was then left until the thermocouple reading had stopped falling before any sweep was started, so that the mount was at a settled temperature and not still on its way down.

Temperature was read with a type K thermocouple bonded to the sample mount and logged through the run; the twelve readings are in Appendix B. The bead is on the mount rather than on the 2DEG itself, and the sample sits in the vapour above the liquid rather than in it, which is why the mean sits a few kelvin above the boiling point rather than at it, and why the temperature is quoted with a generous uncertainty in Section 4.7.

One change to the electrical procedure is needed when cold. The mobility rises by more than an order of magnitude on cooling, so the sample resistance falls and the same bar supply voltage drives a larger current, which risks the 100 µA software cap. Following the script, the bar supply was therefore raised by hand while the current was watched, to find the highest supply voltage that keeps the current inside the cap, before the automated sweep was started. In the event the cold sweep ran over the same 3.0 V to 27.0 V range as at room temperature and reached 90.11 µA at the top, just inside the cap. That the current barely changes on cooling is the 300 kΩ of protection resistance doing its work: it is far larger than the sample, so the drive is close to a current source and the current is set by the supply voltage rather than by the sample.

3.6 Manual familiarisation measurement

The script asks for a manual stage before anything is automated [1]: a small number of readings taken by hand, from which $R_{H,2D}$, R_{sq}, n_{sq} and μ are to be worked out. Its purpose is familiarisation. It confirms that the wiring is right, that V_H and V_L respond to the magnet and to the bar supply in the expected directions and by roughly the expected amounts, and that the numbers that come out are of the right order, before an hour is spent on automated sweeps that would be worthless if any of that were wrong.

Readings of V_H, V_L and I were therefore taken at the top magnet setting and at four bar supply voltages spread across the range, and $R_{H,2D}$ and R_{sq} were worked out from each point on its own, with no fitting and no offset model. Section 4.2 reports what they give and how far that is from the full analysis.

4 Results and analysis

4.1 Electromagnet calibration

The magnet supply was stepped from 0 V to 24 V in 1 V steps into a coil of 10.2 Ω, giving coil currents from 0 to 2.353 A, and the sweep was then reversed. The uncertainty assigned to one calibration point combines a Hall probe repeatability of 2.5 mT, the coil current reading uncertainty of 5.0 mA folded in through the local slope, and a branch reproducibility term of 2.0 mT, giving 3.25 mT to 3.36 mT across the range. The full data set is in Appendix A and is plotted in Figure 1. A weighted

least squares fit of $B(I_m) = c_3 I_m^3 + c_2 I_m^2 + c_1 I_m + c_0$ to the rising branch gave the coefficients in Table 1, together with the comparison against polynomial order.

Table 1: Representative data. Left: cubic calibration of the electromagnet, weighted least squares fit to the rising branch, $\chi^2 = 17.23$ on 21 degrees of freedom and an RMS residual of 2.74 mT against an assumed point uncertainty of about 3.3 mT. Right: goodness of fit against polynomial order.

Coeff.	Value	Std. error	Unit
c_3	+0.004361	0.002414	T A^{-3}
c_2	−0.034490	0.008665	T A^{-2}
c_1	+0.202574	0.008707	T A^{-1}
c_0	+0.002563	0.002335	T

Order	χ^2/ν	ν	RMS (mT)
1, line	8.05	23	8.98
2, quadratic	0.93	22	2.99
3, cubic	0.82	21	2.74

The constant term $c_0 = (2.6 \pm 2.3)$ mT is consistent with zero on its own and cannot by itself be called a remanent field. The evidence for remanence is instead the separation of the two branches extrapolated to zero coil current, 2.56 mT rising against 8.92 mT falling, a difference of 6.4 mT that no single-valued calibration can produce.

A straight line is clearly inadequate, at $\chi^2/\nu = 8.05$: it over-reads the adopted cubic at the top of the sweep by 14 mT, 4 per cent of the calibrated field there, and that error would propagate directly into $R_{H,2D}$ and hence into n_{sq}. The quadratic, however, already describes the data, at $\chi^2/\nu = 0.93$ on 22 degrees of freedom, and going to the cubic buys only $\Delta\chi^2 = 3.3$ for one extra parameter. The fitted $c_3 = 0.004361 \pm 0.002414$ is correspondingly only 1.8 standard errors from zero, so the third order is not demanded by the data. The cubic is retained as the more flexible parameterisation of a curve known on physical grounds to bend over, and the choice costs nothing: the two models differ by at most 2.5 mT at the five working set points, inside the uncertainty on B at every one of them.

The hysteresis is visible in the lowest panel of Figure 1: the mean separation of the two measured branches is 5.52 mT, the two fitted cubics are separated by 6.35 mT at zero coil current, and extrapolated to zero coil current the rising branch gives 2.56 mT against 8.92 mT for the falling branch. A hysteresis loop closes at its turning points, so the branch gap should be largest at low coil current and fall towards zero at the top of the sweep, and the measured branch difference behaves that way. The difference of the two fitted cubics does not: it reaches 8.72 mT at $I_m = 0.71$ A, above its 6.35 mT value at zero current, which is an artefact of differencing two independently fitted cubics rather than a feature of the loop. The measured mean separation of 5.52 mT is the figure carried into the systematic budget of Section 5.

The five working set points used for every later measurement are listed in Table 2. The peak field of 0.3451 T is typical of a teaching-scale iron-cored electromagnet.

Table 2: Representative data. The five working field settings, with the field from the rising-branch cubic and the uncertainty on B at each set point. That uncertainty combines the fit covariance, which alone would give 1.0 mT to 2.3 mT, with the branch-reproducibility term and the coil-current reading error folded in through the local slope. The coil currents are the nominal supply voltage divided by the 10.2 Ω coil resistance; the measured currents in Appendix A differ from these by up to 8 mA, which shifts the calibrated field by under 1 mT and is well inside the quoted uncertainty.

Magnet supply (V)	6.0	11.0	16.0	20.5	24.0
Coil current (A)	0.588	1.078	1.569	2.010	2.353
B from the cubic (T)	0.1107	0.1864	0.2523	0.3058	0.3451
σ_B (mT)	2.47	2.35	2.39	2.36	3.09

4.2 The manual familiarisation measurement

The readings taken by hand in Section 3.6 are in Table 3. They are not a separate data set. They are four rows lifted out of the same representative arrays that Appendices C and D list in full, taken at the top field setting of 0.3451 T, and they are worked here the way they would be worked at the bench: one point at a time on a calculator, with no fitting and no offset term.

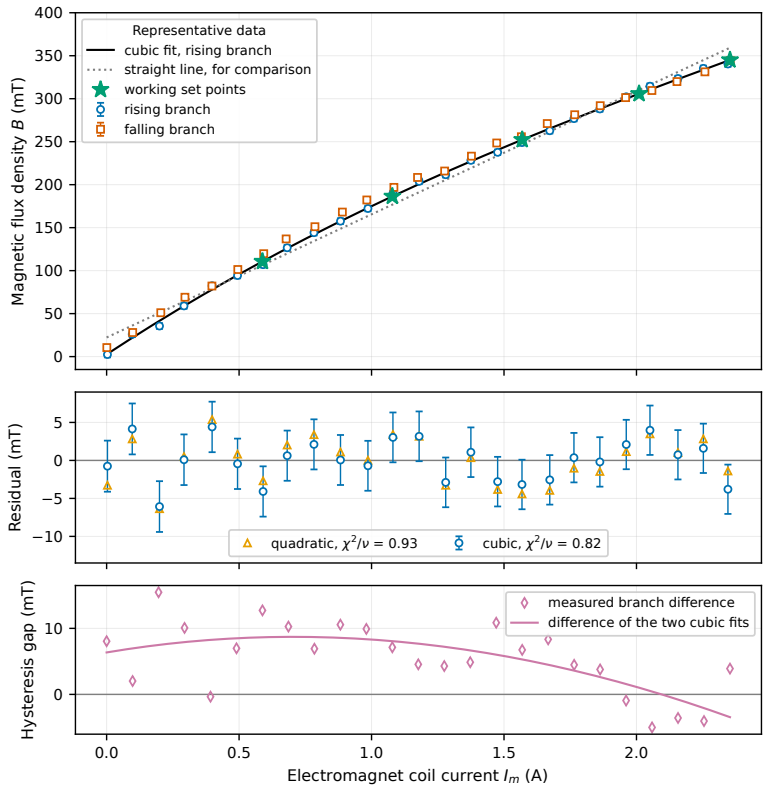


Figure 1: Representative data. Electromagnet calibration against a Hall probe magnetometer. Top: both hysteresis branches with the cubic fit to the rising branch, a straight line for comparison, and the five working set points marked. Middle: residuals of the quadratic and cubic fits. Bottom: the gap between the two branches, measured and from the difference of the two fitted cubics.

Table 3: Representative data. The manual familiarisation readings, a four-point subset of the arrays in Appendices C and D, reduced point by point as $R_{H,2D} = V_H/(IB)$ and $R_{sq} = V_L/(3I)$. The two currents differ slightly between the Hall and longitudinal columns because they come from two separate sweeps at the same bar supply voltage, one with the magnet on and one with it off.

Bar supply	Hall pair, 0.3451 T		Longitudinal pair, zero field			
(V)	I (µA)	V_H (mV)	I (µA)	V_L (mV)	$R_{H,2D}$ ($\Omega\,T^{-1}$)	R_{sq} ($\Omega/\square$)
3.000	9.84	4.558	9.73	46.18	1343	1583
10.059	33.04	14.806	33.18	154.89	1299	1556
18.529	60.87	26.897	60.66	284.37	1281	1563
27.000	88.51	39.110	88.56	415.85	1280	1565
Mean					1301	1567

Averaging the four gives $R_{H,2D} = 1301\,\Omega\,T^{-1}$ and $R_{sq} = 1567\,\Omega/\square$, and equation (7) then gives

$$n_{sq} = \frac{1}{e\,R_{H,2D}} = 4.8 \times 10^{15}\,m^{-2}, \qquad \mu = \frac{R_{H,2D}}{R_{sq}} = 0.83\,m^2\,V^{-1}\,s^{-1}, \tag{14}$$

quoted to two figures, because two figures is all this method supports. Against the full-array results of Table 7 the density is 5.1 per cent low and the mobility 4.9 per cent high.

That error is not random, and it would be wrong to quote the four points as though it were. The standard error on the mean of the four Hall coefficients is $15\,\Omega\,T^{-1}$, about 1.1 per cent, but the four values fall steadily from $1343\,\Omega\,T^{-1}$ at the lowest current to $1280\,\Omega\,T^{-1}$ at the highest, which is a trend and not scatter, so the standard error on the mean measures nothing real. Section 4.5 identifies the trend. Dividing the full model through by IB leaves $V_H/(IB) = m_H + k/B + c/(IB)$, so a point-by-point ratio carries a bias of $k/B + c/(IB)$, and at this field setting that bias runs from $124\,\Omega\,T^{-1}$ at the lowest current down to $51\,\Omega\,T^{-1}$ at the highest, 10 to 4 per cent of the Hall

coefficient. Taking the mean of those four bias terms off the manual average leaves $1227\,\Omega\,\mathrm{T}^{-1}$, within 0.7 per cent of the $1234.9\,\Omega\,\mathrm{T}^{-1}$ adopted later, so the bias accounts for essentially all of the discrepancy.

The longitudinal side behaves much better, and it should: R_{sq} from the four points is 0.4 per cent from the fitted value of Section 4.3, because the longitudinal pair carries no offset term and the measurement has no field in it at all.

The honest summary is that the manual stage gets n_{sq} and μ right to about five per cent. That is enough to confirm the wiring, the polarity and the order of magnitude, which is what it is for, and it is not enough for anything else. It cannot properly be quoted with an uncertainty, because four points sharing one systematic give no way to estimate one. Turning it into a result with a defensible error bar is what the 90-point arrays and the three-parameter fit of Section 4.5 are for.

4.3 Resistance per square at room temperature

The first objective is answered by a single zero-field sweep. The longitudinal voltage between contacts 1 and 4 was recorded against sample current with the magnet off; an extract is given in Table 4, the sweep is plotted in Figure 2 and the full 18-point data set is in Appendix C. A weighted straight line $V_L = m_L I + c_L$ gives $m_L = R_L = (4682.8 \pm 6.1)\,\Omega$ with an intercept of $(0.36 \pm 0.30)\,\mathrm{mV}$, consistent with zero, and $\chi^2/\nu = 0.95$. Dividing by the three squares,

$$R_{\mathrm{sq}}(300\,\mathrm{K}) = \frac{m_L}{3} = (1560.9 \pm 2.0)\,\Omega/\square \ \text{(statistical)}, \qquad (1561 \pm 16)\,\Omega/\square \ \text{(total)}. \tag{15}$$

Table 4: Representative data. Extract from the zero-field longitudinal sweeps at both temperatures: every third point, together with the last point of the sweep. The complete tables are in Appendix C.

Bar supply	300 K		79 K	
(V)	I (µA)	V_L (mV)	I (µA)	V_L (mV)
3.000	9.73	46.18	10.07	3.58
7.235	23.72	111.91	24.08	8.63
11.471	37.51	176.39	38.36	13.64
15.706	51.56	241.63	52.07	18.59
19.941	65.63	306.64	66.18	23.52
24.176	79.24	371.88	80.34	28.50
27.000	88.56	415.85	89.81	31.86

4.4 Dependence of the Hall voltage on current and field

Equation (4) makes a sharp prediction: V_H should be a single linear function of the *product* IB, not separately of I and of B. The test was made in two stages. At 300 K the bar supply was swept over 18 steps at each of the five field settings, giving currents from 9.69 µA to 88.75 µA, Hall voltages from 1.804 mV to 39.110 mV and products IB from 1.073 µA T to 30.543 µA T. The cryogenic array covered 9.71 µA to 90.11 µA, 1.710 mV to 43.875 mV and 1.075 µA T to 31.033 µA T. Both full arrays are in Appendices D and E.

The polarity is worth recording, because it is half of the objective. The transverse voltage on the Hall pair kept one and the same sign at every field setting and every current, and with the field in the direction marked on the magnet and the current driven from the positive end contact, that sign is the one equation (3) requires for a negative carrier charge. Conduction is therefore by electrons, as the heterostructure requires. The tabulated voltages, and every Hall coefficient quoted below, are magnitudes.

A weighted straight line $V_H = s_j I + b_j$ was fitted at each field setting. Table 5 gives the results at both temperatures and Figure 3 shows the room-temperature family. Every line is straight within the scatter, which answers the current half of the objective.

Fitting the five slopes against field as $s(B) = m_H B + k$ gives $m_H = (1241.0 \pm 3.4)\,\Omega\,\mathrm{T}^{-1}$ with $k = (13.08 \pm 0.79)\,\Omega$ at 300 K and $m_H = (1414.3 \pm 3.5)\,\Omega\,\mathrm{T}^{-1}$ with $k = (1.75 \pm 0.80)\,\Omega$ at 79 K. The

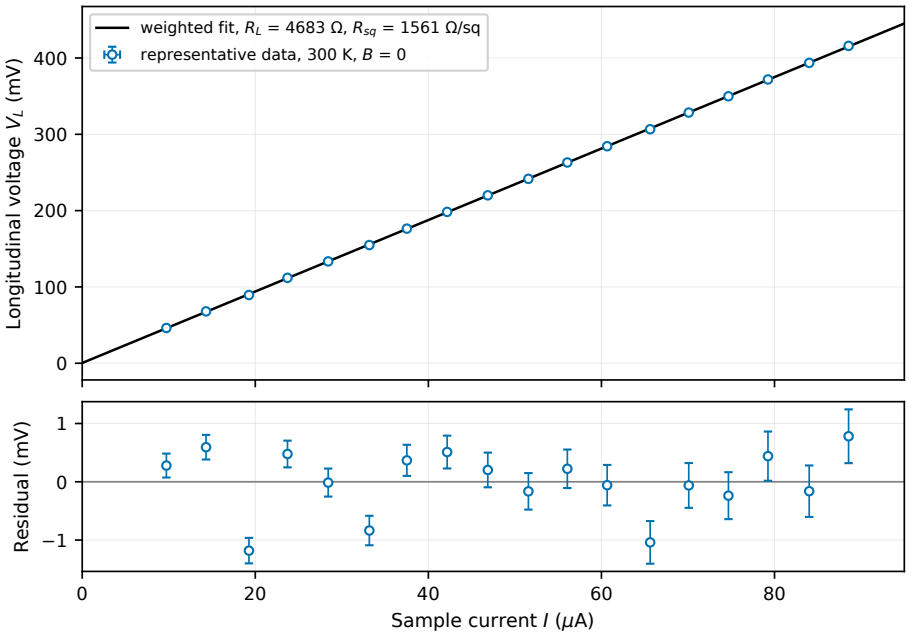


Figure 2: Representative data. Longitudinal voltage against sample current at zero field and 300 K, with the weighted straight-line fit and its residuals. The fitted slope is $R_L = 4683\,\Omega$, giving $R_{\mathrm{sq}} = 1561\,\Omega/\square$.

Table 5: Representative data. Weighted straight-line fits of V_H against I at each field setting and at both temperatures. The uncertainties on the intercepts are 0.029 mV to 0.041 mV throughout. The room-temperature intercepts are all positive and average 0.27 mV, about four times the cryogenic average of 0.07 mV, and neither set grows with field. In the model adopted in Section 4.5 the intercept at fixed field identifies the current-independent offset c on the Hall pair, which is distinct from the current-proportional misalignment term k: that term is proportional to I and so is absorbed into the fitted slopes s_j, not into the intercepts.

	300 K				79 K			
B (T)	s_j (Ω)	σ_{s_j} (Ω)	b_j (mV)	χ^2/ν	s_j (Ω)	σ_{s_j} (Ω)	b_j (mV)	χ^2/ν
0.1107	149.8	0.5	+0.3220	0.95	157.9	0.5	+0.1080	0.74
0.1864	244.8	0.6	+0.2934	0.71	265.4	0.6	+0.1519	1.19
0.2523	327.3	0.7	+0.2645	2.17	359.9	0.7	+0.0639	1.14
0.3058	393.9	0.7	+0.1862	0.51	434.1	0.8	+0.0481	0.95
0.3451	438.9	0.8	+0.2876	0.92	488.7	0.8	−0.0108	0.96

slopes are linear in B, which answers the field half of the objective. Taken with Table 5, V_H is shown to be a single linear function of the product IB rather than of I and B separately. The χ^2/ν of this second-stage fit, 5.64 and 1.92 on three degrees of freedom, sits above unity because its weights carry only the statistical error on each slope: the five field values come from one shared calibration polynomial, so their errors are correlated rather than independent and are handled in the systematic budget of Section 5 instead of in the weights.

4.5 Reducing the whole array to one number

The final objective asks for the best way of collapsing 90 readings into a single Hall coefficient with a sensible error. Three reductions were compared, at both temperatures, and are set out in Table 6.

R2 forces the contact-misalignment voltage, which is proportional to I and independent of B, into a single constant. That voltage is correlated with the abscissa IB, so it biases the slope. At room temperature R2 returns $1273.8\,\Omega\,\mathrm{T}^{-1}$ against the R3 value of $1234.9\,\Omega\,\mathrm{T}^{-1}$, a difference of 3.15 per cent, and its χ^2/ν of 14.20 says plainly that the two-parameter model does not describe the data. The lower panel of Figure 4 shows why: the R2 residuals are grouped into visible bands by field setting and that structure is largely gone once the kI term is added. The weighted residual RMS falls from 0.2421 mV to 0.0853 mV.

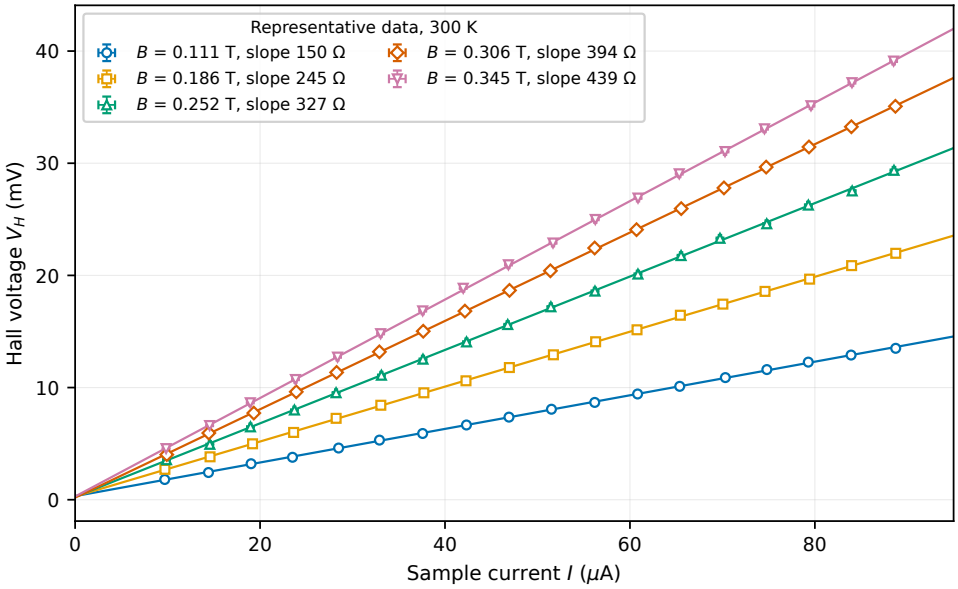


Figure 3: Representative data. Hall voltage against sample current at 300 K for each of the five calibrated field settings, with the weighted straight-line fit at each field.

Table 6: Representative data. Three reductions of the 90-point Hall arrays. $R_{H,2D}$ and its statistical uncertainty are in $\Omega\,T^{-1}$. R1 has three degrees of freedom, R2 has 88 and R3 has 87. R3 is adopted at both temperatures.

		300 K			79 K		
Reduction	Model	$R_{H,2D}$	σ stat	χ^2/ν	$R_{H,2D}$	σ stat	χ^2/ν
R1, slope of slopes	$s_j = m_H B_j + k$	1241.0	3.4	5.64	1414.3	3.5	1.92
R2, naive global line	$V_H = m_H(IB) + c$	1273.8	1.0	14.20	1415.8	1.1	2.41
R3, global with offset	$V_H = m_H(IB) + kI + c$	**1234.9**	**1.5**	**1.51**	**1406.4**	**1.6**	**1.73**

R1 and R3 agree to 0.49 per cent at room temperature and 0.56 per cent cold, so little turns on the choice between them. R3 is preferred because it uses every point in one weighted fit rather than fitting slopes to slopes, and because it returns the offset as an explicit parameter. A fourth possibility, computing $R_{H,2D} = V_H/(IB)$ point by point and taking the mean, is worse than any of the three. Dividing the full model through by IB gives $V_H/(IB) = m_H + k/B + c/(IB)$, so the ratio carries two spurious terms rather than one. It weights the smallest IB points, which have the worst signal-to-noise ratio, equally with the best, and the constant offset contributes $c/(IB)$ to every ratio, a contribution that diverges as IB goes to zero; the misalignment term contributes a further k/B to every ratio, and that one does not fall away as the current is raised. At the lowest field setting it is $130\,\Omega\,T^{-1}$, about 10 per cent of the Hall coefficient itself. That unweighted average returns $1346.2\,\Omega\,T^{-1}$ at room temperature, 9.0 per cent from the adopted value, and $1436.3\,\Omega\,T^{-1}$ cold, 2.1 per cent away.

The nuisance parameters returned by R3 are $k = (14.41 \pm 0.43)\,\Omega$ with $c = (0.277 \pm 0.015)$ mV at room temperature, and $k = (3.44 \pm 0.44)\,\Omega$ with $c = (0.084 \pm 0.015)$ mV cold. The fitted c matches the mean per-field intercept of Table 5 at both temperatures, 0.271 mV and 0.072 mV, which is exactly what this model requires: at fixed B the kI term is proportional to I and so lands in the slope, leaving c alone in the intercept. The quoted standard error on k understates how well it is really known, because k is strongly correlated with m_H: the correlation coefficient from the covariance matrix is -0.75 at room temperature and -0.74 cold. Any error in the field calibration therefore leaks into k as well as into the slope, and k should be read as a parameter that soaks up a field-independent voltage rather than as a precision measurement of the contact geometry.

The objective asks for the density and the mobility, not only for $R_{H,2D}$, so it is worth saying what each reduction would give. At 300 K R1, R2 and R3 give $n_{sq} = 5.03$, 4.90 and $5.05 \times 10^{15}\,m^{-2}$

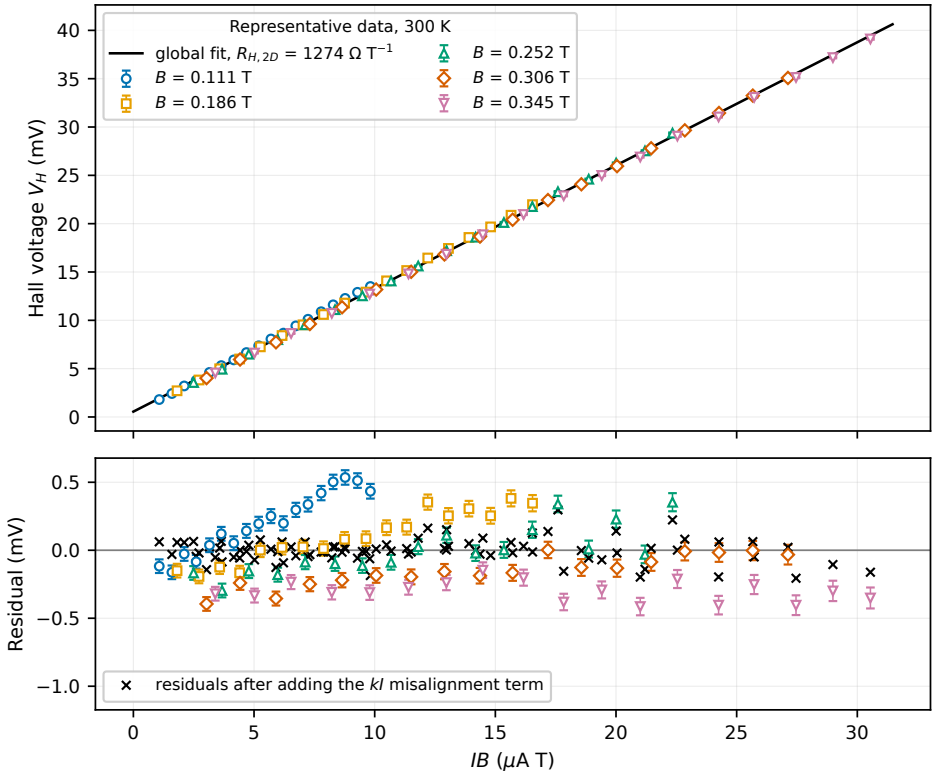


Figure 4: Representative data. Top: all 90 room-temperature Hall readings collapsed onto the product IB, with the naive two-parameter global fit drawn through them. Bottom: residuals of that fit coloured by field setting, together with the residuals of the three-parameter fit as black crosses. The banding by field setting in the first set, and its absence in the second, is the evidence for the contact-misalignment term.

with $\mu = 0.795$, 0.816 and 0.791 m^2 V^{-1} s^{-1}; cold they give $n_{\mathrm{sq}} = 4.41$, 4.41 and 4.44×10^{15} m^{-2} with $\mu = 12.00$, 12.01 and 11.93 m^2 V^{-1} s^{-1}. The total uncertainty is 3.9 per cent on n_{sq} and 4.0 per cent on μ whichever reduction is used, since it is dominated by the magnet calibration and not by the fit, so the choice matters only through the 3.15 per cent bias that R2 carries at room temperature. No comparable choice arises on the longitudinal side: that is one zero-field sweep with no nuisance parameter in it, so m_L enters μ unambiguously.

4.6 Angular dependence of the Hall voltage

The angular scan was taken at 300 K with the magnet supply fixed at 20.5 V, a calibrated field of 0.3058 T, and the bar supply fixed at 18.3 V, giving a mean sample current of 60.07 µA. The rotation stage was stepped through a full 360° in 15° steps; the data are in Appendix F and Figure 5. Fitting equation (10) by weighted least squares gave

$$V_{H0} = (22.855 \pm 0.016)\,\mathrm{mV}, \qquad \theta_0 = (6.07 \pm 0.04)^\circ, \qquad C = (1.002 \pm 0.011)\,\mathrm{mV}, \tag{16}$$

with $\chi^2 = 28.34$ on 22 degrees of freedom, $\chi^2/\nu = 1.29$, and an RMS residual of 0.0582 mV against an assumed meter scatter of 0.050 mV. The excess above unity has an obvious source. The fit holds V_{H0} constant, but the sample current was not perfectly constant across the scan: it ranged from 59.92 µA to 60.44 µA, a standard deviation of 0.2 per cent, which is about 0.05 mV near the peak and so is comparable with the residual itself.

The uncertainties in equation (16) are the fit standard errors, and the fit weights V_H alone. Section 5.2 adds the uncertainty of setting the angle itself. It leaves V_{H0} and C untouched but widens θ_0 to $(6.07 \pm 0.08)^\circ$, and that is the figure quoted elsewhere in this report.

Four things follow. First, the scan is a check on the Hall coefficient: the amplitude predicted from the separately fitted $R_{H,2D}$ and the measured I and B is 22.69 mV against a fitted 22.855 mV, a difference of 0.75 per cent. That difference should not be measured against the 3.9 per cent magnet

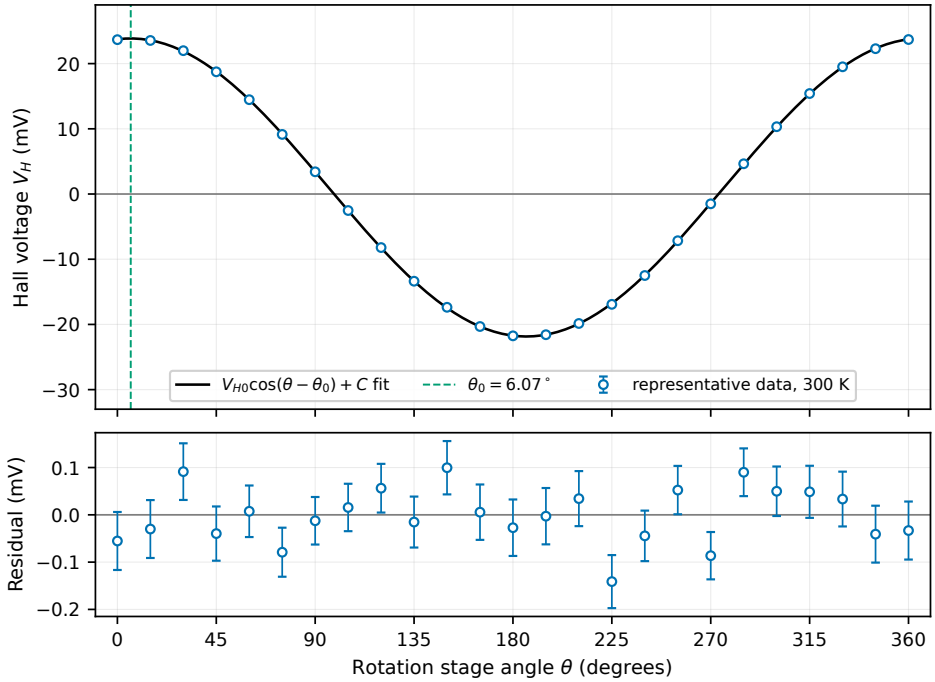


Figure 5: Representative data. Hall voltage against rotation stage angle over a full 360° at 300 K, at fixed field 0.3058 T and fixed mean current 60.07 µA, with the fitted cosine and its residuals. The fitted misalignment $\theta_0 = 6.07°$ is marked.

systematic, because the prediction multiplies $R_{H,2D}$ by the same calibrated field that was used to obtain $R_{H,2D}$ in the first place, so a scale error in B cancels between the two determinations and the comparison tests the shape of the calibration polynomial at one set point rather than its scale. Against the statistical errors that do apply, 0.07 per cent on the fitted amplitude and about 0.12 per cent on the prediction through m_H and the mean current, 0.15 per cent combined, the difference is about five standard errors. The likely causes are the local error of the cubic at this one set point, drift of the current noted above, and the fact that the offset absorbed by the two fits is not the same quantity. The check still bounds any gross error in $R_{H,2D}$ below one per cent.

Second, θ_0 is a mounting misalignment between the zero of the rotation stage and the orientation in which the sample normal is truly parallel to the field, with contributions from the angular scale zero, from how the plug-in box seats in its holder and from the chip not being mounted perfectly flat. It says nothing about the sample, and because the cosine is flat near its maximum a measurement taken at a nominal zero under-reads the normal field component by only 0.56 per cent. Third, the offset $C = 1.002$ mV is field-independent and is close in size to the offset found independently in the global fit, which predicts $kI + c = 1.143$ mV at this current, made up of 0.866 mV from kI and 0.277 mV from c. The two differ by −12.3 per cent, which is about four standard errors once the 0.030 mV on $kI + c$ and the 0.011 mV on C are combined, so they are not the same quantity within their fitted precision. The angular scan measures the offset at one current and one field setting, while k and c are averaged over the whole 90-point array and k correlates with the slope at −0.75. Taken with the temperature dependence of k in Section 6.3, the reading that fits both observations is that a second field-independent term is present which the two-term model $kI + c$ does not fully capture.

Fourth, the fitted cosine term reverses sign at $\theta_0 + 180°$ with equal magnitude while the constant C does not, so the measured voltage does not. The readings at 0° and 180° are 23.674 mV and −21.752 mV: they differ in magnitude by 1.92 mV, which is $2C$ within the scatter, an asymmetry of 8.5 per cent. That asymmetry is itself the signature of the offset. Taking half the difference of two opposite angles, $\frac{1}{2}\,[V_H(\theta) - V_H(\theta + 180°)] = 22.713$ mV, removes C exactly and returns the pure Hall term, while half their sum, 0.961 mV, returns C and so measures the offset a second, independent way. This is the same odd-in-B, even-in-B argument as reversing the field in Section 6.5, and the reversal itself confirms that the signal is a genuine Hall voltage rather than an instrumental artefact.

4.7 Cryogenic run and summary of results

The type K thermocouple readings taken through the cold run, listed in Appendix B, have a mean of 78.69 K, a standard deviation of 1.11 K and a standard error on the mean of 0.32 K. The accuracy floor of a type K junction below 200 K is about 2.5 K and dominates, so the temperature is quoted as $T = (79 \pm 3)$ K. That is a few kelvin above the liquid nitrogen boiling point of 77.4 K at one atmosphere, which is expected: the bead is on the mount rather than on the 2DEG, and the sample sits above the liquid in a pumped dewar. The sensitivity of a type K junction falls from about $41\,\mu\mathrm{V\,K^{-1}}$ at 300 K to roughly $16\,\mu\mathrm{V\,K^{-1}}$ at 77 K, which is why the cold reading is the less precise one. Neither n_{sq} nor μ is calculated from T, so this uncertainty does not propagate into the results; it matters only for comparison with literature values quoted at a nominal 77 K.

The zero-field longitudinal sweep at this temperature gave $m_L = R_L = (353.63 \pm 0.69)\,\Omega$ with an intercept of (0.111 ± 0.035) mV and $\chi^2/\nu = 0.78$, so $R_{\mathrm{sq}} = (117.88 \pm 0.23)\,\Omega/\square$ statistical and $(117.9 \pm 1.2)\,\Omega/\square$ in total. Applying equation (8) at both temperatures gives the results collected in Table 7. Figure 6 places the two temperatures side by side.

Table 7: Representative data. Headline results. $R_{H,2\mathrm{D}}$ comes from the R3 global weighted fit and R_{sq} from the zero-field longitudinal fit. Note that $10^{15}\,\mathrm{m^{-2}}$ is exactly $10^{11}\,\mathrm{cm^{-2}}$, so the density carries the same digits in the units the literature usually quotes. The mobility in those units is $(7910 \pm 320)\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$, the figure carried into Table 10, and $(119\,300 \pm 4800)\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. The product $n_{\mathrm{sq}}\mu$ is equivalently $1/(eR_{\mathrm{sq}})$, in $\mathrm{C^{-1}\,\Omega^{-1}}$. The two ratios between the temperatures, and the changes on cooling, carry statistical uncertainties only: both columns were measured with one field calibration and one value of l/w, so the magnet and geometry systematics are common to them and cancel in any ratio between them.

Quantity	300 K	(79 ± 3) K	Change on cooling
$R_{H,2\mathrm{D}}$, statistical only ($\Omega\,\mathrm{T^{-1}}$)	1234.9 ± 1.5	1406.4 ± 1.6	
$R_{H,2\mathrm{D}}$, total ($\Omega\,\mathrm{T^{-1}}$)	1235 ± 48	1406 ± 54	+13.9 per cent
R_L (Ω)	4682.8 ± 6.1	353.63 ± 0.69	
R_{sq}, total ($\Omega/\square$)	1561 ± 16	117.9 ± 1.2	factor 13.2 decrease
n_{sq}, total ($10^{15}\,\mathrm{m^{-2}}$)	5.05 ± 0.20	4.44 ± 0.17	−12.2 per cent
n_{sq}, statistical only	5.054 ± 0.006	4.438 ± 0.005	
$n_{\mathrm{sq}}(79\,\mathrm{K})/n_{\mathrm{sq}}(300\,\mathrm{K})$	0.8781 ± 0.0015		systematics cancel
μ, total ($\mathrm{m^2\,V^{-1}\,s^{-1}}$)	0.791 ± 0.032	11.93 ± 0.48	factor 15.08 ± 0.04
μ, statistical only	0.7912 ± 0.0014	11.93 ± 0.03	
$n_{\mathrm{sq}}\mu$ ($10^{15}\,\mathrm{V^{-1}\,s^{-1}}$)	3.999 ± 0.040	52.95 ± 0.54	free of magnet error

4.8 Secondary quantities

Several further quantities follow with no extra measurement, taking the electron effective mass in the GaAs conduction band as $m^* = 0.067\,m_0$ [9]. They are collected in Table 8 because the discussion needs them.

The first of them, the momentum relaxation time τ, needs a line of derivation, because it does not follow from anything written down so far. In the Drude picture a carrier is accelerated by the applied field and loses its drift momentum on average once every τ, so in the steady state the momentum gained between collisions balances the momentum lost. Writing magnitudes, as everywhere in this report,

$$\frac{m^* v_d}{\tau} = eE \quad \Longrightarrow \quad \mu \equiv \frac{v_d}{E} = \frac{e\tau}{m^*} \quad \Longrightarrow \quad \tau = \frac{\mu\, m^*}{e}. \tag{17}$$

The mobilities of Table 7 therefore give $\tau = 0.301$ ps at 300 K and 4.55 ps at 79 K. It is τ that then carries into the mean free path $\ell = v_F\tau$ and into the product $\omega_c\tau = \mu B$ in the rows below, both of which the discussion uses.

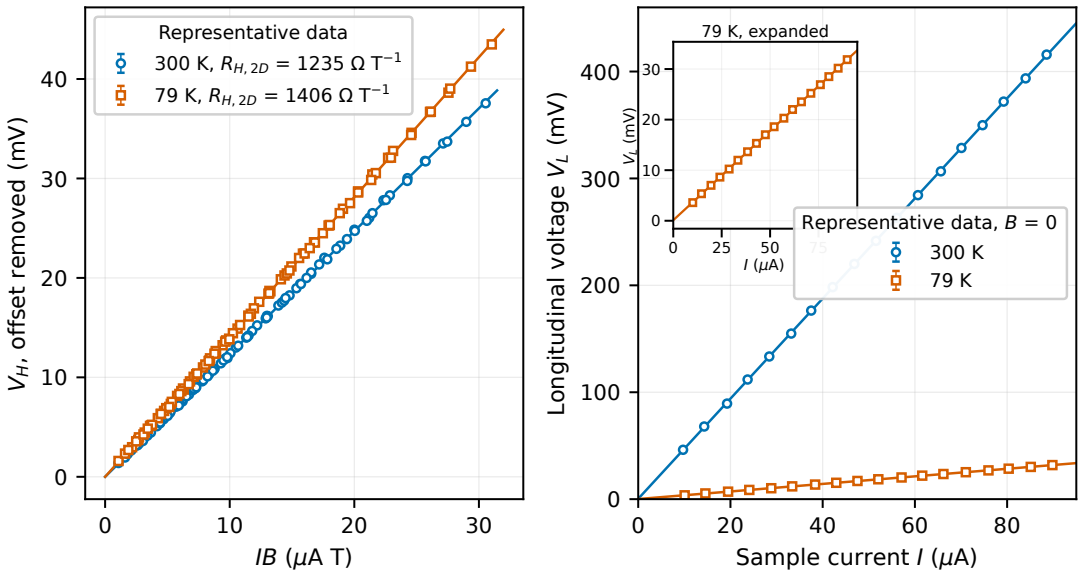


Figure 6: Representative data. Left: Hall voltage against IB at both temperatures with the fitted misalignment term kI and the constant c subtracted, so the remaining signal is the pure Hall term and the fitted lines pass through the origin. Right: raw zero-field longitudinal voltages at both temperatures, with the cryogenic data repeated in an expanded inset.

Table 8: Representative data. Secondary quantities derived from the fitted sheet density and mobility, computed with $m^* = 0.067\,m_0$. The Fermi quantities are zero-temperature expressions. At 300 K the gas is not degenerate, $T/T_F = 1.43$, so the room-temperature entries in the E_F, k_F, v_F and mean free path rows are indicative only: using the thermal velocity $\sqrt{2k_BT/m^*} = 3.7 \times 10^5$ m s^{-1} in place of v_F changes the room-temperature mean free path to 0.11 µm and does not affect the conclusion that transport is diffusive. In particular the room-temperature E_F is a degeneracy scale and not the chemical potential, which is 0.28 meV there; T_F is defined from E_F and is used in this report only to say how far from degeneracy the gas is. At 79 K the chemical potential is 15.16 meV against $E_F = 15.86$ meV, so the two are close and the zero-temperature expressions are nearly right.

Quantity	Formula	300 K	79 K	Unit
Momentum relaxation time τ	$\mu m^*/e$	0.301	4.55	ps
Fermi wavevector k_F	$\sqrt{2\pi n_{\mathrm{sq}}}$	1.782	1.670	10^8 m^{-1}
Fermi velocity v_F	$\hbar k_F/m^*$	3.079	2.885	10^5 m s^{-1}
Mean free path ℓ	$v_F\tau$	0.093	1.311	µm
Fermi energy E_F	$\pi\hbar^2 n_{\mathrm{sq}}/m^*$	18.06	15.86	meV
Fermi temperature T_F	E_F/k_B	210	184	K
$\omega_c\tau$ at 0.345 T	μB	0.273	4.12	dimensionless
Density of states g_{2D}	$m^*/\pi\hbar^2$	2.799	2.799	10^{13} eV^{-1} cm^{-2}

5 Uncertainties

5.1 Random contributions and how the fits were weighted

The random scatter on a single reading is set by the resolution and accuracy of the bench multimeters, by Johnson noise in the sample and leads, by mains pickup on unshielded banana leads, and by slow drift of thermal electromotive forces. The values assigned were a floor of 0.05 mV on V_H with 0.15 per cent of reading in quadrature, a floor of 0.20 mV on V_L at room temperature with 0.10 per cent of reading, a floor of 0.05 mV on V_L at 79 K with 0.15 per cent of reading, and 0.10 µA on I with 0.10 per cent of reading. In relative terms the cryogenic longitudinal measurement is the hardest, because V_L there is only a few tens of millivolts.

Every fit was weighted by $1/\sigma_i^2$ with the abscissa uncertainty folded in through the local slope, so that for the Hall fits $\sigma_i^2 = \sigma_{V_H}^2 + (m_H\,\sigma_{(IB),i})^2$. The magnet calibration uncertainty was deliberately excluded from the weights. It is one correlated systematic shared by every point taken at a given

field setting, not an independent random error on each point, and treating it as random would both mis-weight the fit and understate the final uncertainty. It is carried into the budget below instead. This is why a χ^2/ν of exactly one is not expected for the global Hall fits, and why the adopted R3 values sit at 1.51 and 1.73.

5.2 The angle setting in the angular scan

The list above assigns a scatter to every reading the fits weight. The angular scan of Section 4.6 has a second measured quantity which the fit does not weight at all: the angle itself. The rotation stage was set and read to the nearest degree, which is the resolution at which the scan is recorded in Appendix F. Treating that as a rectangular distribution of half width 0.5° gives a standard uncertainty of $\sigma_\theta = 0.5^\circ/\sqrt{3} = 0.29^\circ$, or 5.0×10^{-3} rad.

Where that matters follows from the derivative of the fitted cosine,

$$\frac{\mathrm{d}V_H}{\mathrm{d}\theta} = -V_{H0}\sin(\theta - \theta_0), \tag{18}$$

which vanishes at the peak. Near $\theta = \theta_0$ the effect is therefore second order, $\Delta V_H \simeq \frac{1}{2} V_{H0}(\Delta\theta)^2$, which for σ_θ is 0.29 µV, a fraction 1.3×10^{-5} of the amplitude. On the flanks, where $|\sin(\theta - \theta_0)| = 1$, it is first order and much larger: $V_{H0}\sigma_\theta = 0.115$ mV, about twice the assigned meter scatter of 0.050 mV to 0.061 mV. The two are equal where $|\sin(\theta - \theta_0)| = 0.48$, about 29° either side of the peak and of the trough, so over most of a full turn it is the angle and not the voltmeter that limits the reading.

Three things follow, and the first two of them are that it does not matter.

It does not touch V_{H0}, which is the only quantity from this scan that any other result depends on. The amplitude is a peak value, the effect there is second order, and 1.3×10^{-5} is four orders of magnitude below the 3.9 per cent magnet systematic. The 0.75 per cent cross-check of the Hall coefficient in Section 4.6 is therefore unaffected, and so is the 0.56 per cent under-reading quoted there for a measurement taken at a nominal zero.

A common error in the zero of the angular scale does not matter either, and for a sharper reason: it is exactly degenerate with θ_0. Adding the same offset to every recorded angle shifts the fitted θ_0 by that offset and leaves V_{H0} and C numerically unchanged. That is the same statement as the one made in Section 4.6, that θ_0 absorbs the angular scale zero along with the mounting, and it is why no systematic on the scale zero appears in the budget below.

What does matter is the random part of the setting error, and it matters for θ_0 itself. A per-point scatter of 0.29° over 25 points gives $\sigma_\theta/\sqrt{N} = 0.06^\circ$, which in quadrature with the 0.04° fit error is 0.07°. Refitting the scan with the angle term folded into the weights, so that $\sigma_i^2 = \sigma_{V_H}^2 + (V_{H0}\sin(\theta_i - \theta_0)\,\sigma_\theta)^2$, gives 0.08° directly, and the larger of the two is adopted: $\theta_0 = (6.07 \pm 0.08)^\circ$. That refit is used to size the uncertainty and not to replace the fitted value; it moves θ_0 by 0.03°, well inside the widened error. The 0.04° of equation (16) is therefore about twice as precise as the stage can justify.

One caution against pushing this any further. Folding the full 0.29° into the weights drops χ^2/ν from 1.29 to 0.44, which says that the scan does not in fact scatter by that much, and that setting a stage against marked graduations is more repeatable than the reading resolution on its own suggests. The true per-point angular scatter is somewhere below 0.29°, so $(6.07 \pm 0.08)^\circ$ is a conservative statement rather than a tight one.

5.3 Propagation

For a quantity $f(x_1, \ldots, x_N)$ formed from independent measured quantities [14],

$$\sigma_f^2 = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_{x_i}^2. \tag{19}$$

From $n_{\mathrm{sq}} = 1/(e m_H)$, $\partial n_{\mathrm{sq}}/\partial m_H = -1/(e m_H^2)$, so the fractional uncertainties are equal in magnitude and

$$\frac{\sigma_{n_{\mathrm{sq}}}}{n_{\mathrm{sq}}} = \sqrt{\left(\frac{\sigma_{m_H}}{m_H}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2_{\mathrm{syst}}}. \tag{20}$$

The field term appears because B enters the abscissa of every point in the fit, so a fractional error in the calibration produces the same fractional error in the slope. From $R_{\mathrm{sq}} = m_L/(l/w)$ and $\mu = (l/w)\, m_H/m_L$, both of which carry the geometry through the ratio l/w,

$$\frac{\sigma_{R_{\mathrm{sq}}}}{R_{\mathrm{sq}}} = \sqrt{\left(\frac{\sigma_{m_L}}{m_L}\right)^2 + \left(\frac{\sigma_l}{l}\right)^2 + \left(\frac{\sigma_w}{w}\right)^2}, \tag{21}$$

$$\frac{\sigma_\mu}{\mu} = \sqrt{\left(\frac{\sigma_{m_H}}{m_H}\right)^2 + \left(\frac{\sigma_{m_L}}{m_L}\right)^2 + \left(\frac{\sigma_l}{l}\right)^2 + \left(\frac{\sigma_w}{w}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2_{\mathrm{syst}}}. \tag{22}$$

R_{sq} carries no magnet term at all, because it was measured at zero field, and n_{sq} carries no geometry term at all, for the reason given after equation (4).

5.4 Systematic contributions

The magnet calibration is by far the largest systematic, and the three terms that make it up have to be kept distinct so that nothing is counted twice. The Hall probe magnetometer has an absolute accuracy of a few per cent, and the probe and the sample do not occupy exactly the same point in a gap where the field is not perfectly uniform; these were combined as 3.0 per cent. The uncertainty on the calibrated field at a set point, listed in Table 2, is 1.2 per cent of B on average; that term already carries the scatter of the calibration points about the cubic, through the fit covariance, together with the branch reproducibility and the coil-current reading. A floor term of 5 mT, which is 2.1 per cent over the mean working field of 0.240 T, covers what the calibration cannot see at all: the branch offset between the rising-branch calibration and the state the core is actually left in by the measurement protocol of Section 4.1, measured as 5.52 mT on average and 6.35 mT extrapolated to zero coil current, together with any drift of the magnet between the calibration sweep and the measurement runs. The scatter about the cubic is not repeated here.

The geometrical systematic enters only through l/w. The mask dimensions are disturbed by lithography and etch bias, typically well under a micrometre, and by lateral depletion at the etched mesa edge, where surface states pin the Fermi level at the exposed sidewall and deplete the 2DEG over a few tenths of a micrometre on each side. Together these were assigned 1.0 per cent. The full budget is in Table 9.

5.5 Correlations, and the quantities that escape the systematic

n_{sq} and μ both contain m_H, so their errors are correlated rather than independent. In the product they cancel exactly:

$$n_{\mathrm{sq}}\, \mu = \frac{1}{e\, R_{H,\mathrm{2D}}} \cdot \frac{R_{H,\mathrm{2D}}}{R_{\mathrm{sq}}} = \frac{1}{e\, R_{\mathrm{sq}}}. \tag{23}$$

The product is therefore fixed by the zero-field longitudinal measurement alone and is completely immune to the magnet calibration. It is known to 1.0 per cent at both temperatures, far better than either factor separately, which is why it is quoted as a line of its own in Table 7.

Any ratio between the two temperatures escapes the same systematics, and for the same reason. Both mobilities are built from the same calibration polynomial and the same l/w, so those factors cancel identically in $\mu(79\,\mathrm{K})/\mu(300\,\mathrm{K})$, leaving only the four fit standard errors of Table 9, 0.29 per cent in quadrature, hence 15.08 ± 0.04. The density ratio behaves the same way: it is $R_{H,\mathrm{2D}}(300\,\mathrm{K})/R_{H,\mathrm{2D}}(79\,\mathrm{K}) = 0.8781 \pm 0.0015$ on statistics alone, so the fall in density on cooling is significant at many standard errors and is not an artefact of the calibration, even though each density separately carries 3.9 per cent.

Table 9: Representative data. Uncertainty budget. The magnet calibration dominates everything that depends on the Hall coefficient, and improving the voltmeters would not change the result. The three magnet terms are kept distinct so that nothing is counted twice: the scatter of the calibration points about the cubic and the branch reproducibility sit in the 1.2 per cent line, the branch offset between calibration and measurement sits in the floor term, and neither appears in both. The angle-setting term is listed separately below the rule because it reaches nothing in the columns above it: it is second order at the peak, so it does not touch V_{H0}, n_{sq}, R_{sq} or μ, and a common scale zero error is absorbed into θ_0 by construction. The per-point setting scatter alone would put 0.06° on θ_0; refitting with the angle folded into the weights gives 0.08°, and that is the figure listed here and quoted throughout. Section 5.2 works it through.

Contribution	Size	Affects
Fit standard error on m_H, 300 K / 79 K	0.13 / 0.11 per cent	n_{sq}, μ
Fit standard error on m_L, 300 K / 79 K	0.13 / 0.19 per cent	R_{sq}, μ
Hall probe absolute accuracy and positioning in the gap	3.0 per cent	n_{sq}, μ
Floor term, 5 mT over a mean field of 0.240 T: branch offset and magnet drift	2.1 per cent	n_{sq}, μ
Calibrated field at a set point: fit covariance, branch reproducibility and coil-current reading	1.2 per cent	n_{sq}, μ
Combined magnet systematic on B	**3.9 per cent**	n_{sq}, μ, not R_{sq}
Geometry, l/w from etch bias and mesa edge depletion	1.0 per cent	R_{sq}, μ, not n_{sq}
Combined on n_{sq}	**3.9 per cent**	both temperatures
Combined on R_{sq}	**1.0 per cent**	both temperatures
Combined on μ	**4.0 per cent**	both temperatures
Rotation stage set and read to 1°, $\sigma_\theta = 0.29°$: second order at the peak, first order on the flanks	0.08° on θ_0	θ_0 only

Two limitations are not included in the budget and are stated here. The Hall scattering factor has been taken as exactly one throughout. At 79 K, $T/T_F = 0.43$ from Table 8, so the gas is partly degenerate and taking $r_H = 1$ is a reasonable approximation. At 300 K, $T/T_F = 1.43$, so the gas is not degenerate, the energy averaging that produces r_H does not disappear, and r_H may exceed unity by several per cent. Since the Boltzmann result quoted in Section 2.1 is $R_{H,\mathrm{2D}} = r_H/(n_{\mathrm{sq}}e)$, so that $n_{\mathrm{sq}} = r_H/(eR_{H,\mathrm{2D}})$, that would make the true room-temperature density a few per cent *higher* than quoted, not lower. This pushes in the opposite direction to the second limitation, parallel conduction in the doped AlGaAs layer at room temperature, which raises the apparent density and lowers the apparent mobility and freezes out on cooling, so the two unquantified effects partly cancel.

6 Discussion

6.1 Comparison with accepted values

Table 10 sets each result against published values for comparable GaAs/AlGaAs modulation-doped structures.

The mobilities agree well. At room temperature the result sits 1.1 per cent from the value reported for the closest published structure, comfortably inside the 4.0 per cent uncertainty. It also sits 6.9 per cent below the bulk undoped GaAs figure, which is outside that uncertainty, but that comparison is not one that should agree exactly: bulk GaAs is a different transport problem, with the carriers and the ionised donors in the same volume, and a 2DEG mobility slightly below the best bulk value is what polar optical phonon limiting predicts. At 79 K the result sits 2.9 per cent above one published value and 2.0 per cent above the original demonstration of Störmer *et al.* [6].

That agreement should be expected, and the reason is worth stating. At room temperature the mobility of a GaAs 2DEG is limited by polar optical phonon scattering. GaAs is a polar crystal, so the longitudinal optical phonon carries a macroscopic electric field that couples strongly to electrons, and its energy of about 36 meV is comparable with $k_BT = 25.85$ meV at 300 K, so that channel is wide open [8]. The phonon population is set by temperature and not by the wafer, so every good GaAs 2DEG has roughly the same room-temperature mobility. At 79 K, k_BT has fallen to 6.81 meV,

Table 10: Representative data. Comparison of the derived quantities with published values for comparable structures. The wafer used in the teaching laboratory is not identified, so exact agreement is not expected for the density. The R_{sq} row is not a directly published figure: it is $1/(n_{\mathrm{sq}} e \mu)$ evaluated from the same comparison structure's own quoted $n_{\mathrm{sq}} = 5.6 \times 10^{11}\,\mathrm{cm}^{-2}$ and $\mu = 8000\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, so that like is compared with like.

Quantity	This work	Literature	Difference
n_{sq}, 300 K ($10^{11}\,\mathrm{cm}^{-2}$)	5.05 ± 0.20	5.6 [10]	−9.7 per cent
μ, 300 K ($\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$)	7910 ± 320	8000 [10]	−1.1 per cent
n_{sq}, 79 K ($10^{11}\,\mathrm{cm}^{-2}$)	4.44 ± 0.17	4.3 [10]	+3.2 per cent
μ, 79 K ($\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$)	119300 ± 4800	116000 [10]	+2.9 per cent
μ at 77 K ($\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$)	119300 ± 4800	117000 [6]	+2.0 per cent
R_{sq}, 300 K ($\Omega/\square$)	1561 ± 16	1393, implied [10]	+12.0 per cent
μ, 300 K ($\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$)	0.791 ± 0.032	0.85, bulk GaAs [12]	−6.9 per cent
$\mu(\mathrm{cold})/\mu(300\,\mathrm{K})$	15.08	14.5 [10]	+4.0 per cent

far below the optical phonon energy, and that channel is largely frozen out, leaving residual acoustic phonons together with remote and background impurity scattering and interface roughness. The 77 K mobility is therefore still partly phonon-limited and close to universal for good material. Only at 4.2 K, where phonons are gone entirely and mobilities above $100\,\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ are reported [1], does the number become a measure of wafer purity.

The room-temperature sheet density is 9.7 per cent below the published figure, larger than the 3.9 per cent uncertainty. That is not surprising, because the density is a design parameter of the particular wafer, set by the doping level, the aluminium fraction and above all the spacer thickness, and the teaching-laboratory wafer is not identified. A thicker spacer gives higher mobility and lower density, so a sample with a slightly lower density than the comparison structure and a comparable mobility is self-consistent. The resistance per square is consistent with that reading. Since $R_{\mathrm{sq}} = 1/(n_{\mathrm{sq}} e \mu)$, a density 9.7 per cent below the comparison structure with a mobility 1.1 per cent below it must give an R_{sq} about 12 per cent *above* it, and it does: the comparison structure's own n_{sq} and μ imply $R_{\mathrm{sq}} = 1393\,\Omega/\square$, against the $1561\,\Omega/\square$ found here. The round figure of about $2000\,\Omega/\square$ often quoted for HEMT material is a generic value for a different wafer, is not consistent with the n_{sq} and μ of the comparison structure, and is worth no percentage at all; it serves only as an order-of-magnitude check.

6.2 What the change on cooling shows

The clearest single result is the contrast in Table 7. On cooling, R_{sq} falls by a factor of 13.2 while $R_{H,\mathrm{2D}}$ changes by only 13.9 per cent. That asymmetry is the direct fingerprint of modulation doping, because the two quantities are controlled by different physics. The sheet density is fixed by charge transfer equilibrium across the heterojunction, which involves the donor binding energy, the conduction band offset and the electrostatics of the transferred dipole, all energies of tens to hundreds of millielectronvolts. Changing $k_B T$ from 25.85 meV to 6.81 meV barely disturbs that balance, so the density is almost temperature-independent. Because both temperatures share one magnet calibration and one l/w, the 3.9 per cent systematic cancels in the ratio between them, exactly as it does in the mobility ratio and in $n_{\mathrm{sq}}\mu$: the density ratio is 0.8781 ± 0.0015 on statistics alone, so the fall is significant at many standard errors and is not an artefact of the calibration. The fall of 12.2 per cent that is observed is the normal size and has two plausible causes: thermal de-trapping of a small fraction of electrons at room temperature, and freeze-out of any parallel conduction in the doped AlGaAs layer, which at room temperature contributes a spurious extra carrier population to a Hall measurement. The mobility, in contrast, is set by scattering rates, and the dominant room-temperature channel is exponentially sensitive to temperature through the phonon occupation number. A conventionally doped semiconductor, where every carrier is accompanied by an ionised donor in the same volume, cannot behave this way, because impurity scattering does not freeze out. The measured mobility ratio is 15.08 ± 0.04, where again the uncertainty is statistical only because the calibration and the geometry cancel in the ratio, and it sits close to the 14.5 of the structure it

most resembles [10].

The high mobility is also visible in the raw readings, before any analysis. At 79 K the Hall voltage at the top of the sweep, 43.9 mV, is larger than the longitudinal voltage, 31.9 mV; at room temperature the reverse holds, 39.1 mV against 415.8 mV. The condition for the inversion is simply $\mu B > l/w$, and from Table 8 $\mu B = 4.12$ at the peak field when cold against $l/w = 3$, while at room temperature $\mu B = 0.273$.

6.3 Why room temperature is the harder measurement

Section 2.4 predicted that the contact-misalignment offset scales as $IR_{\mathrm{sq}}\delta/w$, so that it is much worse at room temperature than cold. The direction is confirmed three separate ways. The fitted misalignment term k rises from 3.44 Ω cold to 14.41 Ω at room temperature. The bias from ignoring the offset, the difference between R2 and R3, is 3.15 per cent at room temperature and 0.67 per cent cold. The error made by naively averaging point-by-point ratios is 9.0 per cent at room temperature and 2.1 per cent cold. The per-field intercepts of Table 5 cannot be used as a fourth test, for the reason given with that table: at fixed B the misalignment term is absorbed into the fitted slope, so the intercept measures the current-independent offset c and not the current-proportional one.

The size, however, is not confirmed. The three ratios above are 4.2, 4.7 and 4.3, all close to four, whereas equation (11) makes the offset strictly proportional to R_{sq} and so demands 13.2. Inverting that equation with the fitted k sharpens the point: $\delta = w\,k/R_{\mathrm{sq}}$ gives an apparent contact displacement of 0.92 µm at 300 K but 2.92 µm at 79 K on a 100 µm bar. A lithographic displacement is fixed when the bar is defined and cannot change with temperature, so two values a factor of three apart are direct evidence that contact misalignment is not the only source of the field-independent voltage. Part of the cold offset must come from something else that does not scale with R_{sq}, most plausibly a thermal electromotive force on the Hall pair, whose leads span a gradient of more than 200 K when the sample is cold; and part of the discrepancy is that the fitted cold k is inflated by its correlation of -0.74 with the slope, which lets field-calibration error leak into it. The practical conclusion is unchanged: the room-temperature measurement, which looks like the easy one because the signals are larger and no cryogen is involved, is the one in which an uncorrected offset produces the larger error.

6.4 Limits of the classical treatment

At 79 K and 0.345 T the product $\omega_c\tau = 4.12$, which in principle satisfies the condition for Landau quantisation to be resolved, so it is worth checking whether Shubnikov-de Haas oscillations should have been visible. The Landau level spacing at that field is $\hbar\omega_c = \hbar eB/m^* = 0.596$ meV, against $k_BT = 6.81$ meV. Thermal broadening exceeds the level spacing by a factor of eleven, so the oscillations are completely washed out, the classical analysis is valid, and no quantum Hall features are expected. It should be said that these data could not have shown such oscillations in any case: the Hall array is five discrete field settings rather than a continuous field sweep, and R_{sq} was deliberately measured at zero field only, so there is no trace of longitudinal resistance against field to look at. The thermal broadening argument is what establishes that none are expected. Two further assumptions can be checked against Table 8. The mean free path is 0.093 µm at room temperature and 1.311 µm cold, both far shorter than the 300 µm probe separation, so transport is diffusive and the Drude picture applies. The two-dimensional density of states, $2.799 \times 10^{13}\,\mathrm{eV^{-1}\,cm^{-2}}$, is independent of energy, which is the most characteristic signature of two-dimensionality and is what makes the sheet density, rather than a volume density, the right variable throughout.

6.5 Improvements to the measurement

The budget in Table 9 shows where effort would be repaid. The magnet calibration contributes essentially all of the 3.9 per cent uncertainty on n_{sq}. Improving the voltmeters would not measurably reduce that uncertainty; only a better field measurement would. Three concrete steps would help: use a magnetometer probe with a traceable calibration and a stated absolute accuracy; mount it in a jig that places its active element at the exact position the 2DEG occupies, so that gap non-uniformity

is removed rather than estimated; and calibrate immediately before and after each measurement run rather than once, so that drift is bounded.

Measuring at $+B$ and $-B$ and taking $V_H = \tfrac{1}{2}[V_H(+B) - V_H(-B)]$ would remove the offset exactly, because the Hall term is odd in B and the offset is even. Reversing the current as well would remove thermal electromotive forces, which are independent of the sign of I while the Hall voltage reverses with it. That matters most when cold, where leads run from the cold finger to a room-temperature connector across a gradient of more than 200 K and any asymmetry between the two leads of a pair produces a drifting offset of tens to hundreds of microvolts. The present analysis fits the offset instead, which works, but at the cost of a parameter that correlates with the slope at -0.75.

Three smaller changes would also help. Demagnetising the core with a decreasing alternating current before each calibration and each measurement, rather than relying on always approaching from the same direction, would put calibration and measurement on the same branch at last. That would remove the branch offset now carried inside the 5 mT floor term, and the 2.0 mT branch reproducibility from the uncertainty on each calibration point. Replacing the type K thermocouple on the mount with a calibrated silicon diode or platinum sensor on the chip carrier would tighten the quoted (79 ± 3) K and would allow $\mu(T)$ to be measured over a range of temperatures and fitted to a power law, which identifies the dominant scattering mechanism directly. And five field settings is the minimum for a convincing test of linearity in B; ten or twelve over the same range would make any residual curvature from an imperfect calibration polynomial visible in the residuals rather than absorbed into the slope.

7 Conclusion

The Hall coefficient and the resistance per square of a GaAs/AlGaAs two-dimensional electron gas were determined at room temperature and under liquid nitrogen cooling. The Hall voltage was confirmed to be a single linear function of the product of sample current and normal flux density, and its angular dependence followed $V_{H0}\cos(\theta - \theta_0) + C$ with a mounting misalignment of $\theta_0 = (6.07 \pm 0.08)^\circ$, the amplitude agreeing with the value predicted from the current and field sweeps to 0.75 per cent, which is about five times the statistical uncertainty of that comparison.

Adopting a three-parameter global weighted fit that separates the Hall term from a field-independent contact-misalignment term, the results are $R_{H,2\mathrm{D}} = (1235 \pm 48)\,\Omega\,\mathrm{T}^{-1}$ and $R_{\mathrm{sq}} = (1561 \pm 16)\,\Omega/\square$ at 300 K, and $(1406 \pm 54)\,\Omega\,\mathrm{T}^{-1}$ and $(117.9 \pm 1.2)\,\Omega/\square$ at (79 ± 3) K. These give a sheet carrier density of $(5.05 \pm 0.20) \times 10^{15}\,\mathrm{m}^{-2}$ falling to $(4.44 \pm 0.17) \times 10^{15}\,\mathrm{m}^{-2}$, a change of -12.2 per cent, and a mobility of $(0.791 \pm 0.032)\,\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ rising to $(11.93 \pm 0.48)\,\mathrm{m}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, a factor of 15.08 ± 0.04 in which the magnet and geometry systematics cancel, leaving statistics alone. Both mobilities agree with published values for comparable structures to within 3 per cent and the mobility ratio to within 4 per cent.

The near-constant carrier density set against a mobility that rises by more than an order of magnitude is the expected signature of modulation doping, in which the carrier concentration is fixed by charge transfer equilibrium and the mobility is limited by phonon scattering that freezes out on cooling. The dominant uncertainty throughout is the calibration of the iron-cored electromagnet, at 3.9 per cent, and the product $n_{\mathrm{sq}}\mu = 1/(eR_{\mathrm{sq}})$, which is measured at zero field and is therefore immune to it, is known to 1.0 per cent at both temperatures.

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A Electromagnet calibration data

Table 11: Representative data. Electromagnet calibration against a calibrated Hall probe magnetometer, sample stand removed from the gap. The falling branch was taken from 24 V downwards and is listed here in ascending supply voltage so the two branches can be compared row by row. The uncertainty on each field reading is 3.25 mT to 3.36 mT.

Rising branch			Falling branch		
Supply (V)	I_m (A)	B (mT)	Supply (V)	I_m (A)	B (mT)
0	0.0031	2.44	0	0.0008	10.49
1	0.0967	25.98	1	0.0984	28.01
2	0.1996	35.58	2	0.2045	51.04
3	0.2921	58.98	3	0.2957	69.05
4	0.3982	82.44	4	0.3975	82.10
5	0.4942	94.33	5	0.4952	101.31
6	0.5906	106.98	6	0.5930	119.71
7	0.6816	126.62	7	0.6776	136.89
8	0.7827	144.18	8	0.7863	151.10
9	0.8828	157.57	9	0.8899	168.13
10	0.9864	172.30	10	0.9821	182.21
11	1.0807	189.74	11	1.0844	196.87
12	1.1796	203.86	12	1.1737	208.41
13	1.2799	211.59	13	1.2759	215.88
14	1.3749	228.30	14	1.3771	233.17
15	1.4759	237.64	15	1.4725	248.50
16	1.5678	249.02	16	1.5655	255.76
17	1.6727	262.76	17	1.6632	271.09
18	1.7640	276.88	18	1.7668	281.35
19	1.8618	288.11	19	1.8631	291.89
20	1.9614	302.21	20	1.9590	301.27
21	2.0509	314.55	21	2.0582	309.55
22	2.1565	323.50	22	2.1528	319.92
23	2.2524	335.29	23	2.2583	331.26
24	2.3452	340.40	24	2.3540	344.31

B Cryogenic thermocouple readings

Table 12: Representative data. Type K thermocouple readings taken on the sample mount through the cryogenic run, in the order they were logged. The mean is 78.69 K, the standard deviation 1.11 K and the standard error on the mean 0.32 K, but the accuracy floor of a type K junction below 200 K, about 2.5 K, dominates all three, which is why the temperature is quoted as (79 ± 3) K throughout.

Reading	T (K)	Reading	T (K)
1	77.53	7	78.17
2	77.50	8	79.31
3	77.31	9	80.22
4	78.50	10	78.89
5	79.17	11	80.46
6	77.48	12	79.67

C Zero-field longitudinal data

Table 13: Representative data. Longitudinal voltage between contacts 1 and 4 against sample current, magnet off, at both temperatures. The assigned uncertainties are 0.21 mV to 0.46 mV on V_L at 300 K, 0.050 mV to 0.069 mV at 79 K, and 0.10 µA to 0.13 µA on the current.

Bar supply	300 K		79 K	
(V)	I (µA)	V_L (mV)	I (µA)	V_L (mV)
3.000	9.73	46.18	10.07	3.58
4.412	14.31	67.95	14.56	5.34
5.824	19.27	89.39	19.49	7.00
7.235	23.72	111.91	24.08	8.63
8.647	28.42	133.45	28.84	10.22
10.059	33.18	154.89	33.39	11.96
11.471	37.51	176.39	38.36	13.64
12.882	42.17	198.33	42.95	15.31
14.294	46.87	220.02	47.54	17.01
15.706	51.56	241.63	52.07	18.59
17.118	56.06	263.09	57.11	20.28
18.529	60.66	284.37	61.59	22.00
19.941	65.63	306.64	66.18	23.52
21.353	70.09	328.50	71.04	25.21
22.765	74.67	349.80	75.95	26.91
24.176	79.24	371.88	80.34	28.50
25.588	84.01	393.59	85.13	30.17
27.000	88.56	415.85	89.81	31.86

D Room-temperature Hall array

Table 14: Representative data. Complete room-temperature Hall array, 90 points. Currents I are in µA and Hall voltages V_H in mV. The field at the head of each pair of columns is the value returned by the rising-branch calibration cubic of Table 1.

Bar supply	0.1107 T		0.1864 T		0.2523 T		0.3058 T		0.3451 T	
(V)	I	V_H	I	V_H	I	V_H	I	V_H	I	V_H
3.000	9.69	1.804	9.71	2.710	9.88	3.564	9.91	4.021	9.84	4.558
4.412	14.42	2.426	14.59	3.826	14.57	4.940	14.46	5.946	14.55	6.615
5.824	19.03	3.210	19.19	4.987	18.96	6.495	19.32	7.726	18.94	8.646
7.235	23.51	3.784	23.63	6.000	23.72	7.997	23.92	9.621	23.83	10.720
8.647	28.49	4.609	28.23	7.257	28.22	9.536	28.28	11.350	28.38	12.717
10.059	32.91	5.316	33.06	8.424	33.12	11.100	32.90	13.185	33.04	14.806
11.471	37.58	5.905	37.72	9.533	37.62	12.531	37.65	15.022	37.59	16.839
12.882	42.32	6.663	42.27	10.603	42.33	14.069	42.15	16.817	41.97	18.856
14.294	46.92	7.364	46.96	11.785	46.78	15.616	46.97	18.664	46.86	20.950
15.706	51.51	8.068	51.72	12.920	51.45	17.199	51.40	20.409	51.68	22.890
17.118	56.19	8.675	56.29	14.084	56.22	18.600	56.18	22.442	56.25	24.989
18.529	60.83	9.427	60.76	15.151	60.86	20.117	60.71	24.076	60.87	26.897
19.941	65.38	10.109	65.45	16.448	65.53	21.765	65.54	25.953	65.33	29.060
21.353	70.32	10.889	70.05	17.439	69.73	23.306	70.17	27.799	70.28	31.043
22.765	74.81	11.604	74.61	18.574	74.79	24.599	74.73	29.655	74.55	33.072
24.176	79.29	12.268	79.43	19.666	79.29	26.267	79.36	31.450	79.58	35.129
25.588	83.91	12.898	83.97	20.873	84.03	27.530	83.95	33.251	84.01	37.183
27.000	88.74	13.499	88.75	21.970	88.55	29.366	88.71	35.075	88.51	39.110

E Cryogenic Hall array

Table 15: Representative data. Complete cryogenic Hall array at (79 ± 3) K, 90 points. Currents I are in µA and Hall voltages V_H in mV. The same five calibrated field settings were used as at room temperature.

Bar supply	0.1107 T		0.1864 T		0.2523 T		0.3058 T		0.3451 T	
(V)	I	V_H	I	V_H	I	V_H	I	V_H	I	V_H
3.000	9.71	1.710	10.01	2.814	9.88	3.686	10.06	4.374	9.92	4.959
4.412	14.47	2.467	14.57	4.101	14.93	5.363	14.67	6.464	14.92	7.161
5.824	19.59	3.127	19.34	5.266	19.44	7.055	19.49	8.480	19.44	9.471
7.235	24.05	3.872	23.95	6.487	24.24	8.783	24.17	10.560	24.07	11.804
8.647	28.72	4.632	28.73	7.721	29.03	10.496	28.70	12.560	28.89	14.012
10.059	33.35	5.358	33.54	9.078	33.37	12.110	33.54	14.638	33.37	16.316
11.471	38.24	6.112	38.13	10.273	38.19	13.809	38.25	16.606	38.17	18.690
12.882	42.76	6.857	43.17	11.524	42.94	15.496	42.86	18.647	42.87	20.976
14.294	47.65	7.577	47.63	12.892	47.33	17.187	47.69	20.698	47.59	23.242
15.706	52.20	8.399	52.12	13.978	52.27	18.956	52.37	22.727	52.25	25.583
17.118	56.97	9.048	57.03	15.184	57.07	20.526	57.15	24.775	56.97	27.800
18.529	61.55	9.890	61.70	16.508	61.71	22.301	61.59	26.808	61.88	30.146
19.941	66.40	10.619	66.36	17.910	66.74	23.870	66.41	28.902	66.48	32.402
21.353	70.97	11.324	71.17	18.985	71.09	25.592	71.05	30.868	71.18	34.700
22.765	75.77	12.064	75.68	20.206	75.66	27.297	75.58	33.114	75.75	37.057
24.176	80.44	12.825	80.22	21.512	80.47	29.061	80.43	34.948	80.26	39.393
25.588	85.21	13.607	85.18	22.819	85.00	30.782	85.27	37.064	85.09	41.621
27.000	89.94	14.283	89.95	23.978	90.02	32.469	90.11	39.020	89.93	43.875

F Angular scan data

Table 16: Representative data. Hall voltage against rotation stage angle at 300 K, at fixed field 0.3058 T and fixed mean current 60.07 µA. The assigned voltage uncertainty runs from 0.050 mV to 0.061 mV across the scan. The negative readings between 105° and 270° are the reversal of the Hall term discussed in Section 4.6; note that they are not equal in magnitude to the positive readings at the opposite angles, the difference being twice the offset C.

θ (deg)	I (µA)	V_H (mV)	θ (deg)	I (µA)	V_H (mV)
0	60.14	23.6741	195	60.19	−21.5785
15	60.02	23.5506	210	60.11	−19.8540
30	60.13	21.9846	225	60.44	−16.9184
45	60.08	18.7425	240	60.17	−12.4990
60	59.92	14.4670	255	59.95	−7.1624
75	59.94	9.1407	270	60.25	−1.5013
90	59.94	3.4074	285	60.00	4.6396
105	59.94	−2.5290	300	60.08	10.3222
120	60.12	−8.2111	315	60.26	15.4121
135	59.97	−13.3737	330	59.92	19.5092
150	60.04	−17.3712	345	60.11	22.2884
165	59.92	−20.3187	360	60.12	23.6961
180	60.10	−21.7516			